MoS2 doping methods for TENG charge traps

"What MoS2 doping methods are used to create charge traps in TENGs" *Generated by LeapSpace*

MoS_2 charge traps in TENGs are created mainly by **defect engineering, surface functionalization, and composite/layer-by-layer incorporation**, rather than by a single standardized substitutional-doping recipe. The clearest TENG-relevant approaches in the retrieved evidence are **plasma-induced defects**, **vacancy/S-vacancy engineering**, **thiol-based surface charge-transfer doping**, and **MoS_2-based hybrid or interfacial charge-trapping layers** [1] [4] [57] [61] [62] [63] [64] [65].

Methods used to create charge traps in MoS_2 for TENGs

Method	How it creates traps	TENG-relevant role
Plasma defect treatment	Plasma treatment generates localized in-gap states and higher defect density in MoS_2, which act as donor/acceptor trap sites [61]	Increases charge storage capacity and trap density [61]
Sulfur-vacancy engineering	S-vacancies introduce defect states in the bandgap; these states are spatially localized around Mo atoms near the vacancy [66]	Creates stable trap sites for captured charges [66]
Electrochemical / covalent surface functionalization	Direct electrochemical functionalization can generate S–C bonds and vacancy-related electronic structure changes [63]	Tuning of trap states and charge capture behavior [63]
Thiol-based molecular chemisorption	Thiol molecules bind to sulfur vacancies and act as surface charge-transfer dopants, changing carrier density and electrical behavior [62]	Useful for engineering charge handling, though the retrieved paper is not TENG-specific [62]
Layer-by-layer self-assembly / functional-layer doping	MoS_2 is assembled as a dedicated charge-trapping layer in a multilayer TENG architecture [67]	Directly builds a trap layer in the device stack [67]
Hybrid filler/composite embedding	MoS_2 in polymers like PVDF, PVC, Ecoflex, or PVDF-TrFE adds trap states and boosts charge capture through interfacial polarization and roughness [4] [57] [59] [68]	Most practical TENG route for raising output [4] [57] [59] [68]
Monolayer / few-layer engineering	Thin MoS_2 layers act as efficient electron acceptors or charge traps because of their 2D surface area and band structure [1] [64] [69]	Enhances charge trapping and surface charge density [1] [64] [69]

What is most commonly used in practice

The most clearly supported TENG strategy is **not heavy chemical doping**, but **creating trap-rich MoS_2 through defects and interfaces**. In the retrieved studies, MoS_2 improves TENG output by acting as a **charge-trapping layer** or **tribonegative filler**, while defect creation and interfacial engineering raise the number of available trap sites [1] [4] [67]. Plasma treatment and vacancy engineering are the most direct ways to create those traps in the MoS_2 itself [61] [66].

Bottom line

If your goal is **charge traps in a TENG**, the best-supported MoS_2 "doping" route is really **trap engineering**: **plasma/defect creation, sulfur-vacancy control, and interfacial composite design**. Substitutional dopants exist in the broader MoS_2 literature, but the TENG evidence you provided mainly supports **defect-rich MoS_2 and MoS_2/polymer hybrid layers** as the charge-trap mechanism [4] [61] [62] [63] [65].

Confidence level: **High** — multiple TENG-relevant sources consistently support defect engineering and hybrid-layer construction as the main ways MoS_2 is modified to create charge traps, while direct substitutional-doping evidence for TENGs is limited.

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